Flashback 2 X4: research and implementation

A playable study of delay behavior, using the original Braids oscillator engine and an independently implemented Wasp-style filter. The software is useful for exploring sounds and routing; it is not a verified sonic duplicate of a physical Flashback 2 X4.

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1. Scope and evidence

“Flashback 2x4” is interpreted here as TC Electronic’s Flashback 2 X4 Delay. Product generations must be kept separate: a feature description for an original X4, a compact Flashback 2, or a Triple Delay does not automatically establish the behavior of the 2 X4. This implementation models one selected delay algorithm, with stereo internal memory and optional multiple rhythmic taps. It does not expose three independently stackable delay engines.

The primary pedal reference is TC Electronic’s 2019 user manual, hosted as a complete PDF by B&H; the manufacturer’s download endpoint was also located. The manual establishes operations and intended effect categories. It does not supply the DSP algorithms, transfer functions, tuning constants, or measurements needed to prove sample-level equivalence. The official product page confirms the product identity.

No verified service schematic or source release for this exact pedal was located in the reviewed public material. This is a bounded finding, not proof that none exists. General-purpose delay schematics, product descriptions, firmware update packages, and descriptions of other Flashback generations cannot fill that gap. A convincing reproduction would additionally require either the original implementation or controlled input/output measurements from a physical unit.

The oscillator situation is different. Mutable Instruments publishes the DSP code and describes a desktop test path, permitting direct integration rather than imitation. The filter sits between these cases: a Wasp clone circuit is available and its topology is interpretable, but this build does not solve every component’s nonlinear behavior. These different evidence levels are preserved throughout the interface and report. [2]

Subsystem	Evidence available	What this build can claim
Braids oscillator	Original C++ and lookup tables	Actual upstream DSP, with a documented rate-conversion wrapper

Subsystem	Evidence available	What this build can claim
Pedal controls	Manufacturer operation manual	Broadly reproduced interaction model; mapping exceptions listed below
Delay sound	Effect descriptions; no reference measurements	Independent behavioral approximation
Wasp filter	Manufacturer description and published clone schematic	Topology-informed coloration, not a component-accurate solver
Demo patches	Original authored sequences and settings	Reproducible examples of this software

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2. Pedal behavior

The manual specifies nine named delays, six presets in two banks, six TonePrint slots, a typical 20–7000 ms range, additive echo level, rhythmic subdivisions, pressure control, MIDI, and looping. Stereo/mono loop capacities are 20/40 seconds. Buffered bypass permits trails. Crystal raises feedback echoes by an octave; Dynamic ducks them during input. MIDI reception uses channel 6. [1, §§4–8]

The web panel reproduces those concepts with conventional accessible controls. Its blue enclosure is a visual reference rather than a dimensionally exact CAD recreation. Sliders and numeric readings deliberately make settings inspectable. The main Delay control becomes inactive under tempo sync, avoiding a misleading display in which a knob appears to affect a clock-controlled time. Turn sync off before testing tape-time sweeps.

The host adds a synth, audition switches, source gain, output volume, sequencer, file input, and rendered comparisons. Those are laboratory features. They should not be inferred to exist on the pedal. Likewise, a browser’s output level and latency do not describe the analog input headroom, converter latency, noise floor, or impedance of the hardware.

Signal path

BRAIDS × 8 @ 96 kHz → FIR downsample → voice drive → voice SVF → voice VCA → stereo sum

STEREO SUM / AUDIO FILE → unity dry branch ——————————→ SUM → output gain → ceiling
└→ delay memory → algorithm → echo level —┘
↑_____ feedback _____|

This is the implemented signal path, not a traced circuit diagram of the TC pedal. The filter is per voice, so a four-note chord uses four independently enveloped filters. An imported audio file enters at the effects input and bypasses the synth’s voice filter. The output ceiling is a host safeguard for extreme settings; it does not model pedal clipping.

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3. Delay implementation

All delay processing occurs in the same original JavaScript engine used for the live instrument and offline WAVs. The code is in dsp.js. Two 15-second floating-point circular buffers support long delays, fractional reads, reverse playback, and the additional history needed by the pitch shifter. The clean read interpolates between adjacent samples. Memory wraps explicitly, including the floating-point edge case in which a tiny negative position rounds to the buffer length.

At a fixed integer delay D, the clean core follows $w[n] = x[n] + f \cdot w[n - D]$ and the echo branch is $e[n] = L \cdot w[n - D]$. An impulse therefore produces echoes at D, 2D, 3D, with successive amplitude ratio f. The direct branch is added independently. This equation is the implementation’s reference behavior and provides an exact, testable baseline for the other algorithms.

Mode	Implementation in this build	Unverified elements
2290	Uncolored fractional-delay memory and feedback	Converter character, interpolation, exact control curves
Analog	Two cascaded low-pass stages, capped at 2.6 kHz, with mild feedback saturation	BBD clock behavior, companding, noise, hardware filter response
Tape	Feedback saturation, two low-pass stages capped at 5.2 kHz, two sinusoidal speed perturbations, slower time smoothing	Head response, hysteresis, physical transport statistics, factory tuning
Dynamic	Peak follower controlling wet attenuation, with faster attack than recovery	Factory threshold, knee, time constants, and level dependence
2290 Mod	Three differently modulated fractional read taps, averaged; opposite stereo modulation	Actual chorus topology, phase relationships and modulation range
Crystal	Two complementary windowed read heads moving at 2× playback rate inside the feedback path	TC pitch-shifter windowing, diffusion, anti-aliasing, and latency
Reverse	Two overlapping reverse grains with complementary windows	Phrase detection, window scheduling, dry/wet timing conventions

Mode	Implementation in this build	Unverified elements
LoFi	8-bit quantization and a four-sample hold after bandwidth reduction	Actual factory bit depth, sampling behavior, nonlinearities and noise
Ping Pong	Mono sum injected left; cross-feedback alternates successive echoes between channels	Hardware stereo-input handling, pan law and dual-subdivision interactions

Time, subdivisions and feedback

The clock is expressed as quarter-note BPM. An individual tap is calculated as `60000 / BPM × ratio` milliseconds. Ratios are 1.5, 1, 2/3, 3/4, 1/2, 1/3, and 1/4, plus three paired patterns and an editable custom ratio. Each primary delay is bounded to 20–7000 ms. A paired subdivision uses two averaged read taps sharing a feedback buffer. That is a deliberate, inspectable implementation choice; the pedal’s exact paired-tap feedback graph has not been established.

Time smoothing produces a moving read head, and therefore pitch motion during a time change. Tape uses a slower one-pole smoothing coefficient than the other modes. This is musically useful, but it also means non-Tape modes may bend pitch during rapid time edits. No claim is made about the hardware’s transition policy for those modes. Changing algorithms clears the software’s delay memory; seamless cross-algorithm spillover is not implemented.

The front-panel feedback slider spans zero to 0.95. MASH and expression can extend it to 0.985. The cap prevents deliberately unbounded feedback in this instrument and differs from any hardware setting that may self-oscillate. With a clean single tap, a rough amplitude decay estimate is `T60 = D × ln(0.001) / ln(f)`. Colored, shifted, reverse, or multiple-tap algorithms do not necessarily obey that estimate because their processing changes the loop gain and spectrum.

Crystal and reverse timing

Crystal uses a 3840-sample grain window at the 48 kHz effects rate. The two read phases are half a cycle apart, and their raised-cosine weights sum to one. The read delay decreases by one sample per output sample within a grain, so the effective source position advances twice as quickly. Repeating this operation in feedback creates successively higher pitch registers. A steady 400 Hz test verifies a first-pass result near 800 Hz. This is a pitch-direction check, not a perceptual match against TC’s implementation.

The grain window adds up to 80 ms of additional history beyond the selected delay. Consequently, “400 ms” should not be interpreted as a sample-exact onset measurement for Crystal. Reverse similarly uses a selected window length rather than a guaranteed first-echo onset. A reverse grain reads backward through captured history; two staggered grains crossfade to reduce discontinuities. These choices can smear transients, and both modes are explicitly approximations.

Dynamic and coloration controls

The Dynamic follower uses absolute input amplitude, then applies `1 / (1 + 22 × envelope)` to the echo branch. It does not apply that ducking gain inside feedback. Echoes can therefore keep circulating while the output is attenuated and become audible when the input subsides. Voice Gain strongly affects this behavior: increasing it can make the direct synth louder while simultaneously reducing the echo-to-dry ratio during a phrase.

Echo Tone changes the colored algorithms' low-pass setting, subject to their mode caps. It does not secretly filter the clean modes. Wow / Chorus Depth changes Tape speed variation and 2290 Mod tap excursion. The interface identifies those applicability limits. Colored feedback uses soft saturation without artificial random noise, making repeated test renders deterministic and keeping silence silent.

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4. Braids integration

The compiled oscillator files are `analog_oscillator.cc`, `digital_oscillator.cc`, `macro_oscillator.cc`, `resources.cc`, and stmlib's `random.cc`, together with their headers. The original source identifies analog building blocks, digital models, and the macro layer that combines them. [3]

The wrapper creates eight independent MacroOscillator instances. Note numbers are converted to Braids' pitch representation using 128 units per semitone. Timbre and Color are scaled into its parameter range. Percussive and physical models receive a Strike call on note-on. The engine exposes the 47 normal model selections; the hidden "question mark" mode is not included.

The upstream lookup-table generator specifies a 96 kHz oscillator rate. Rendering it directly at 48 kHz would alter pitch and timing. The wrapper therefore renders two native 24-sample blocks for each host block, low-pass filters them with a 63-tap Blackman-windowed sinc, and decimates by two. The host receives 24 samples at 48 kHz. The original oscillator C++ and lookup tables are unchanged. [4]

This preserves the model source while deliberately changing its output stage. It is not a complete Braids hardware replica: there is no DAC, analog reconstruction circuit, control-voltage calibration, front-panel firmware, display, quantizer UI, or hardware signature control. Eight-voice allocation, panning, envelopes and filtering belong to the host. The WebAssembly module has no imported system functions and requires no network access after the page assets are loaded.

For reproducibility, the eurorack revision is `08460a69a7e1f7a81c5a2abcc7189c9a6b7208d4`; stmlib is `d18def816c51d1da0c108236928b2bbd25c17481`. The selected DSP files carry MIT notices. The GPL desktop test program was consulted as a build reference but is not compiled into the instrument. Credits and licensing text accompany the distributed assets.

5. Filter analysis

Doepfer describes the A-124 as a 12 dB multimode design using CMOS inverters as analog amplifiers, with a separate band-pass output and a low-pass/high-pass blend that gives a notch near its midpoint. The published description specifically distinguishes its behavior from self-oscillating filters. [5]

The inspected Jürgen Haible clone drawing shows CD4069UB inverter stages, two CA3080 transconductance stages, capacitor integration, a resonance return with a diode pair, output selection, and an additional distortion path. The drawing is a clone circuit, not a Flashback schematic and not evidence that every A-124 revision uses identical values. René Schmitz's page identifies the circuit and links the original drawing. [6] [7, circuit drawing]

This evidence supports using a two-integrator state-variable structure rather than a four-pole ladder as the starting topology. It also motivates nonlinear coloration. It does not determine the inverter's exact transfer function, transconductance dependence, diode conduction, supply behavior, component tolerances, or the response of an individual unit. A true circuit model would need a nonlinear solver and calibrated component models.

The implemented core uses trapezoidal integration. With `g = tan(πfc/fs)` and damping `k`, it computes `a1 = 1 / (1 + g(g+k))`, `a2 = g·a1`, and `a3 = g·a2`. Low-pass, band-pass and high-pass outputs are derived from the same state update. The numerical approach follows the family of state-variable formulations discussed by Andrew Simper. [8]

Input coloration is an offset hyperbolic tangent with the offset's DC value subtracted. The transfer is asymmetric, producing a different harmonic balance from a perfectly odd saturation curve. A following DC blocker removes accumulated offset. The two integration states are bounded to protect numerical stability, but this is not a solution of the circuit's feedback diode conduction. In normal operation the filter core remains a linear state-variable structure driven by a nonlinear input.

Resonance reduces damping but never makes it negative. Low-pass and high-pass are blended linearly, while band-pass can be selected separately. Filter cutoff is bounded to 25–16000 Hz, with exponential modulation in octaves. A dedicated ADSR controls cutoff independently of the amplitude ADSR. Those controls make the source expressive while keeping the distinction between a Wasp-inspired voice and a component-accurate Wasp emulation explicit.

What a closer filter model would require

A next implementation would identify the exact circuit revision, translate its nodes into equations, model the OTA and inverter transfer curves, and fit parameters against several reference sweeps. Test signals should include low-level sine sweeps, increasing-amplitude tones, two-tone intermodulation tests, and resonance sweeps at several cutoff settings. Matching only a static frequency response would miss the level-dependent character that makes this circuit interesting.

6. Performance controls

Each voice has amplitude and filter ADSRs. Attack is linear, decay approaches sustain exponentially, sustain holds while the gate remains active, and release decreases linearly from the current envelope value. Attack and release can be short enough for percussion or long enough for chord swells. Retriggering starts from the current envelope level. When eight voices are occupied, the oldest is reused; there is no separate crossfade for voice stealing.

Two LFOs provide sine, triangle, or square shapes. Four modulation rows route LFO 1, LFO 2, MASH, or expression to cutoff, Timbre, Color, pitch, pan, or delay time. Amounts are bipolar. Cutoff destinations cover five octaves per unit; pitch covers twelve semitones per unit; delay modulation is exponential in octaves of delay length. These are host modulation ranges, not recovered hardware calibration values.

MASH and expression each also have three dedicated mappings to time, feedback and echo level. The MASH slider represents pressure continuously; holding a mouse-controlled footswitch applies its maximum value. A mouse does not measure foot pressure, and these response curves do not claim to match TC’s factory mappings. Releasing a held switch restores MASH to zero. The keyboard can be played with mouse, touch or the computer keys shown on screen.

Six preset memories are arranged as A/B/C in two banks and persisted to browser storage. A second group of six local recipe slots stores host patches. These recipes are analogous to an editable slot workflow only: the application neither reads nor writes proprietary TonePrint data, beams presets, updates firmware, nor communicates with a pedal over USB. Preset recall retains the source oscillator and envelope/filter settings so the same phrase can be used to compare delay settings.

Web MIDI accepts note input on all channels and handles pedal controls on channel 6. The host interprets the manual’s program-change numbers as transmitted values 1–7; controllers displaying one-based patch labels may show different numbers. MIDI clock is averaged over 48 clock intervals. Continuous-controller scaling and the bypass threshold are host conventions because the reference table does not establish full response curves. MIDI device behavior has not been tested against a physical controller or pedal.

Looper mode replaces the delay engine. The first recording fixes the loop length, later passes add overdubs, and one saved buffer supports undo/redo of the most recent overdub. Play Once stops at the next boundary. Mono is selected before recording to obtain the longer capacity. The loop exists only in memory; refreshing the browser discards it. Audio-file input and live output recording make the laboratory usable with material beyond the included sequencer.

7. Listening programme

The eight compositions are deliberately short and sparse. Each emphasizes a different oscillator and delay interaction, with space after the phrase for echoes to be heard. They are original examples, not transcriptions of existing music or recordings of the pedal. The downloadable wet files contain two pattern cycles plus ten seconds of tail. Dry files use the exact source performance and the same source/output gain, making them useful references without independent normalization.

Session	Source and effect	Suggested listening comparison
Amber current	Saw/sub bass → Tape	Compare repeat brightness; change unsynced time during a sustained tail
Glass transit	FM keys → Ping Pong	Listen on headphones for alternating stereo echoes
Cloud atlas	Swarm chords → Crystal	Stop the source and follow the octave ladder
Copper strings	Physical pluck → Analog	Compare the first pluck’s brightness with later repeats
Vowel engine	Formant sequence → Dynamic	Compare echo prominence during the phrase and its long rest
Backward constellation	Struck bell → Reverse	Contrast direct strikes and reversed swells
Raster garden	Wavetable arpeggio → 2290 Mod	Switch to clean 2290 while retaining the source
Eight-bit comet	Tuned percussion → LoFi	Listen for texture on low-level decays; compare to clean 2290

Open the audio comparison room for synchronized switching between the rendered dry and processed files. For hands-on tests, the instrument’s three audition buttons expose full mix, dry source, and echoes only. Dry audition intentionally clears existing delay memory, so it is a source-isolation tool rather than a seamless hardware bypass test. Use the pedal footswitch and Trails setting to test bypass behavior.

Changing a session loads an entire source/effect patch and stops the sequence. Changing only the delay type keeps the source patch, allowing a more controlled comparison. Use one short pluck or the bell session when examining echo envelopes; long pads make timing differences harder to hear. To examine Dynamic, retain the phrase’s empty second half. To examine Crystal, allow the tail to continue rather than repeatedly restarting the sequence.

8. Validation

Automated validation separates software correctness from hardware fidelity. A passed assertion verifies a property of the implemented engine under a defined test signal. It does not certify equivalence with TC Electronic’s algorithms. The test outputs and render measurements are distributed with the application rather than summarized as an unsupported “sounds identical” claim.

The DSP suite exercises all 47 oscillator models for finite, nonzero output; measures a sustained sine A4; checks sample-exact clean impulse timing and feedback ratio; verifies ping-pong channel alternation; tests unity dry routing, kill-dry, and true/buffered bypass behavior; checks octave-up pitch shifting; stresses every delay mode at maximum feedback plus MASH; and exercises looping, overdub, undo, redo, stop-at-boundary, clear and memory limits. Machine-readable DSP results.

Browser checks run the actual AudioWorklet, start source sequences, exercise audition controls, switch presets and recipes, operate the looper, export a WAV and inspect a narrow viewport for horizontal overflow. The downloadable reference renders record peak level, RMS level and the number of samples reaching the host output ceiling. Browser results · Render measurements and exact patches.

Render	Seconds	Peak dBFS	RMS dBFS	Ceiling samples
Amber current	18.9	-12.02	-35.20	0
Glass transit	18.3	-12.13	-29.67	0
Cloud atlas	23.3	-12.63	-29.29	0
Copper strings	19.6	-12.02	-32.76	0
Vowel engine	18.0	-13.10	-38.28	0
Backward constellation	20.7	-13.36	-36.47	0
Raster garden	17.7	-12.18	-30.20	0
Eight-bit comet	18.6	-12.20	-34.03	0

RMS figures include silent gaps and the ten-second tail and are not LUFS measurements. The files are not independently loudness-normalized. Peak and ceiling counts establish numerical headroom, not subjective quality or absence of intersample peaks. The code can be inspected and the tests rerun; no listening judgment is represented as a measured hardware match.

9. Fidelity limits and next measurements

A complete reproduction remains unproven. The largest unknowns are the pedal’s exact feedback topologies, wet/dry gain curves, modulation trajectories, delay-time transition policy, pitch/reverse windowing, input dynamics, and nonlinear level dependence. TonePrint compatibility and proprietary preset contents are absent. Physical switching behavior, analog dry circuitry, impedance, converter noise and latency, pressure-sensor mechanics, and MIDI THRU hardware are outside this software model.

The Wasp-style filter is also explicitly approximate. Its drive is not the same as modeling the circuit’s inverter, OTA and diode interactions. Braids is the strongest part of the source match because it uses the real DSP, but its host polyphony, FIR output conversion, filter and envelopes create a new instrument rather than a complete replica of the module.

Controlled pedal capture protocol

For a defensible hardware fit, use a fixed audio-interface loop with its direct round-trip reference recorded first. Feed a calibrated impulse, swept sine, steady tones, broadband noise bursts and a few sparse musical phrases. Capture left and right outputs separately. Record firmware version, delay type, all knob positions, subdivision, bypass mode, kill-dry, and exact input level for every take.

At minimum, measure three delay times, three feedback settings and three input amplitudes per algorithm. First establish delay onset and repeat amplitudes from impulses. Then estimate repeat-to-repeat spectral change with sweeps or noise bursts. Use a long sine to measure pitch modulation over time. For Dynamic, alternate loud tone bursts with silence and fit the wet recovery curve. For Crystal, measure each feedback generation’s fundamental, sidebands and onset separately.

Pressure mapping needs its own test: record a slow MASH sweep alongside the control value if available, or use repeatable MIDI values as a separately labelled proxy. Compare bypass tails and preset changes independently. A good fit should be checked on held-out signals, not only on the samples used to tune it. Publish residual timing, level and spectral errors rather than an arbitrary “percentage accurate.”

Until those measurements exist, this laboratory should support creative auditioning and understanding effect families. It should not be the sole basis for judging the exact sound of a pedal purchase. The difference is evidential: the software is fully inspectable, but the reference hardware has not been measured.

10. Reproduction and delivery

The precompiled `public/braids.wasm` is included. The public folder is a static site with no backend, analytics, accounts or paid APIs. Serve it over localhost or HTTPS so AudioWorklet and Web MIDI can run. A simple local launch is `python3 -m http.server 18742 --bind 127.0.0.1 --directory public`. Open the resulting localhost URL, enable audio and select a session.

The source archive includes the host DSP, UI, wrapper, build script, tests, renderer, required upstream DSP sources and license notices. The pinned portable compiler is Zig 0.14.1, used as a C++ compiler targeting wasm32-wasi in reactor mode. The produced module imports no WASI calls. The build command is in `tools/build.sh`; install the matching compiler in its documented location to rebuild. A modern Node installation can run the DSP suite and audio renderer without starting a browser.

Offline WAV export uses a worker and the same engine as live playback. It renders the selected demo pattern and current patch, not a recording of hand-played notes or imported audio. “Record output” captures an actual live performance using the browser’s supported MediaRecorder container. The browser labels the downloaded extension accordingly. Recipe and preset persistence uses localStorage; audio files remain local to the page.

Sources and references

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